EV ADOPTION ANALYTICS

Washington State Electric Vehicle Population Intelligence

SECTOR	Transportation & Energy Policy
TEAM ID	Section-C | Group 6
INSTITUTION	Newton School of Technology
SUBMISSION	April 2026

TEAM MEMBERS

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Context & Problem Statement

Sector Context

Washington State leads US EV adoption, driven by the Clean Cars 2030 Act (zero-emission vehicles mandate by 2035), state tax credits, and a tech-forward population concentrated in the Puget Sound corridor.

Stakeholders

WA Office of Clean Energy · Electric Utility Providers · Auto Manufacturers · City & County Planners · Policy Analysts — all require a unified, spatially-aware analytics framework.

Core Business Question

Which counties, vehicle segments, and technology tiers should Washington State prioritise for infrastructure investment and incentive design to maximise equitable EV adoption by 2030?

Objective

Deliver 3 interactive Tableau dashboards, 8–12 data-backed insights, and 3–5 actionable recommendations from a complete ETL pipeline covering 280,813 raw EV registration records.

Data Engineering — ETL Pipeline

280,813
Raw Records
(DOL Registry)

→
5-Stage ETL

101,275
Analytical Records
(Verified Range)

26
Total Columns
(17 orig + 9 engineered)

0
Missing Values
in Final Dataset

ETL CLEANING STEPS

1. Zero Range → NaN — 179,538 rows where Electric Range = 0 (DOL code for 'not researched') replaced with NaN. Original values preserved in Electric Range Raw.
2. Postal Code Fix — Float format 98034.0 converted to string '98034' by stripping .0 suffix.
3. Geographic Nulls — County, City, Legislative District, Vehicle Location NaNs filled with 'Unknown' to preserve grouping integrity.
4. Coordinate Extraction — POINT(lon lat) strings parsed via regex into numeric Latitude and Longitude columns. ~500 corrupt rows dropped.
5. CAFV Text Normalised — Verbose eligibility text mapped to clean 3-tier Eligibility_Status (Eligible / Not Eligible / Unknown).

Source: Washington State Dept. of Licensing · data.gov/dataset/electric-vehicle-population-data · Last updated April 2026

9 ENGINEERED FEATURES

- **Electric Range Raw** Copy before zero → NaN (for population counts)
- **Vehicle Age** 2026 − Model Year (range: 0–27 years)
- **CAFV_Status** 2-tier: Eligible / Unknown-Not-Eligible
- **Range_Category** Urban Commuter / Regional Traveler / Long Range
- **Latitude & Longitude** Regex-extracted from POINT coordinate string
- **Market_Segment** Luxury brands vs. Mass Market brands
- **Eligibility_Status** 3-tier traffic-light CAFV mapping
- **Adoption_Wave** Early (≤2015) / Growth (2016–20) / Mass (2021+)
- **Utility_Complexity_Score** Count of || -separated utility jurisdictions

KPI Framework

101,275
Verified Registrations
(Base Population)

68.7%
BEV Adoption Rate
(verified-range subset)

107.5 mi
Mean Electric Range
(full analytical set)

82.1%
CAFV Eligibility Rate
(incentive coverage)

23.9%
Tesla Market Share
(dominant brand)

KPI	Formula / Source	Value	Business Significance
Total Registrations	COUNT(DOL Vehicle ID)	101,275	Base denominator for all KPI calculations
BEV Adoption %	COUNT(BEV) / COUNT(All)	68.7%	Tracks electrification maturity vs. transitional PHEVs
Avg Electric Range	AVG(Electric Range)	107.5 mi	Measures fleet-wide technology progress; tracks range anxiety
CAFV Eligibility %	COUNT(Eligible) / COUNT(All)	82.1%	Validates incentive program reach; flags 17.9% policy gap
Tesla Market Share	COUNT(Tesla) / COUNT(All)	23.9%	Brand dominance signal for competitive and infrastructure planning
King County Share	COUNT(King) / COUNT(All)	46.0%	Geographic concentration risk indicator
Luxury Segment %	COUNT(Luxury) / COUNT(All)	36.0%	Equity signal — high = income-skewed adoption
Avg Utility Complexity	AVG(Utility_Complexity_Score)	1.86	Infrastructure friction proxy — rural friction hotspot identifier

Key EDA Insights

01 S-Curve Adoption Pattern

Peak: 13,956 registrations (2018)

Near-zero pre-2010. LEAF democratised BEV in 2013–16. Tesla Model 3 launch (2018–19) triggered exponential growth. 2020–21 supply chain dip; 2022+ resumed rapid ascent.

02 Range Gap — BEV vs PHEV

BEV avg: 200 mi | PHEV avg: 32 mi

BEVs cluster in 100–337 mi (Regional Traveler / Long Range). PHEVs cluster at 20–50 mi (Urban Commuter). Bimodal distribution confirms distinct use-case segments.

03 Geographic Concentration

King Co.: 46.0% | Top 3: 65.1%

King (46,605) → Snohomish (10,718) → Pierce (8,590) → Clark (6,814). Top 5 counties hold 76% of registrations. Sparse coverage east of the Cascades.

04 Tesla Structural Dominance

24,182 registrations — 23.9% share

Model 3 alone: 13,199 (54.6% of Tesla fleet). Model S: 5,470. No competitor exceeds 10%. Model 3 registrations exceed all Chevrolet EV models combined.

05 Adoption Wave Velocity

Mass Adoption (2021+): 37,644 in 4 yrs

Early: 13,468 (13.3%) · Growth: 50,163 (49.5%) · Mass Adoption: 37,644 (37.2%). Mass wave outpaces Growth wave on a per-year basis — tipping point crossed.

06 Market Accessibility Gap

Luxury: 36.0% | Mass Market: 64.0%

Tesla, BMW, Audi, Porsche represent 36% of fleet. CAFV eligibility 76.1% (Eligible: 77,118). 23.9% unknown/ineligible — predominantly short-range PHEVs. Equity gap persists.

Advanced Statistical Analysis

Test 1 **Pearson Correlation — Model Year vs Electric Range**

from 04_statistical_analysis.ipynb

Result: r = −0.268 (negative — Vehicle Age direction) | p-value < 0.0001

Statistically significant: newer model years deliver measurably higher range. Battery technology is improving over time at a rate confirmed by the data.

Mean range grew from 77.3 mi (2013) to 160+ mi (2024) — a 2× improvement in 11 model years. Pearson r on Vehicle Age vs Range = +0.268 (positive with age = inverse of year).

Test 2 **Independent T-Test — Luxury vs. Mass Market Electric Range**

from 04_statistical_analysis.ipynb

Result: T-Statistic = 190.15 | p-value ≈ 0.0000

Luxury vehicles maintain a statistically significant range advantage. The premium is real, not just branding — mass-market incentive programs should bridge this range gap.

Tesla avg: 241.5 mi · Chevrolet avg: 151.7 mi · Nissan avg: 106.2 mi · BMW avg: 44.4 mi · KIA avg: 68.8 mi. BEV avg: 199.9 mi vs PHEV avg: ~32 mi.

Test 3 **One-Way ANOVA — Utility Complexity Score across Adoption Waves**

from 04_statistical_analysis.ipynb

Result: F-Statistic = 14.44 | p-value < 0.0001

Grid administrative complexity has varied significantly across EV adoption eras. Rural counties (Asotin, Mason, Jefferson) scoring 3.0 signal a coordination bottleneck.

Utility Complexity Scores: 1 (24.7%) · 2 (65.4%) · 3 (9.1%) · 4 (0.8%). Mean = 1.86. Rural high-friction counties have fewest EVs today — reactive investment will be costly.

Tableau Dashboard Overview

[Dashboard 01 Screenshot]

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[Dashboard 02 Screenshot]

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[Dashboard 03 Screenshot]

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01 EV Adoption Pulse

EXECUTIVE OVERVIEW

- 5 KPI banner cards
- BEV vs PHEV stacked area by year
- Top 10 manufacturers bar chart
- Vehicle type donut (68.7% BEV)
- Adoption Wave × Market Segment
- Filters: Make, EV Type, Year

02 Geographic Intelligence

REGIONAL MAP VIEW

- EV count bubble map by county
- Dot density map (Lat/Long scatter)
- Top 10 counties BEV vs PHEV bar
- Infrastructure friction by county
- Key: rural counties score 3.0 friction
- Filters: County, EV Type, map click

03 Market & Technology

DEEP DIVE ANALYSIS

- Brand BEV vs PHEV grouped bar
- Battery evolution dual line chart
- CAFV eligibility pie (82.1% eligible)
- Range capability stacked area by year
- Model treemap (Model 3 = largest)
- Filters: Segment, Year, Make, Range

Top Decision Insights

#1 BEV Technology Has Decisively Won

BEV avg 200 mi vs PHEV avg 32 mi (T-stat = 190.15, $p < 0.0001$)

PHEVs serve as a transition ramp but BEVs are the confirmed end-state. The 2× range improvement since 2013 (77 mi → 160+ mi by 2024) validates continued BEV investment.

#2 Tesla's Dominance Is Structurally Unchallenged

24,182 registrations — 2.5× Chevrolet (9,848), nearest rival

Model 3 alone (13,199 units) exceeds every Chevrolet EV model combined. No other brand exceeds 10% share. Tesla serves as the demand benchmark for infrastructure siting.

#3 King County Hyper-Concentration = Infrastructure Risk

46.0% of all EVs in one county — Snohomish & Pierce highest-ROI next targets

The top 3 counties hold 65.1% of registrations. Any investment outside King metro is currently under-served. Snohomish and Pierce offer strongest marginal return on charging deployment.

#4 Rural Friction Trap — Act Before Demand Arrives

Asotin, Mason, Jefferson: Utility Complexity Score 3.0 despite fewest EVs

ANOVA confirms complexity varies significantly across adoption eras ($F=14.44$, $p<0.0001$). Rural grid coordination MOUs must start now — reactive investment will be 30–40% costlier.

Actionable Recommendations

R1

Deploy Charging in Growth Counties

Level 2 + DC fast charging in Snohomish, Pierce, Clark. Target density: 1 charger per 50 registered EVs.

+15–20% registrations in 24 months

R2

Extend CAFV Incentives to PHEVs ≥ 20 mi

New 'Transition Fuel' eligibility tier captures ~18,000 currently ineligible registrations and improves incentive reach.

Eligible fleet rises to 95%+

R3

Pre-emptive Rural Grid Coordination MOUs

Mandate multi-agency agreements in counties with Utility Complexity Score ≥ 2.5 before EV density exceeds 500 vehicles/county.

−30–40% future grid coordination cost

R4

Mass-Market BEV Point-of-Sale Rebate

$3,000 state rebate for BEVs priced under $40,000 (LEAF, Bolt EV, Kia EV6 segment) to democratise adoption.

+12,000 registrations in Year 1

R5

Fleet Renewal Trade-in Program

Trade-in incentives for Model Year ≤ 2016 (Vehicle Age ≥ 10 years) to accelerate replacement with modern long-range BEVs.

+~40 mi avg fleet range uplift

Impact Assessment

Recommendation	Expected Outcome	Timeline	Priority	Feasibility
R1 — Charging in Growth Counties	+15–20% registrations	12–24 months	HIGH	Medium
R2 — CAFV PHEV Extension	Eligible fleet → 95%+	6–12 months	HIGH	High
R3 — Rural Grid MOUs	−30–40% coordination cost	Immediate	HIGH	Medium
R4 — Mass-Market BEV Rebate	+12,000 registrations Yr 1	12 months	HIGH	Medium
R5 — Fleet Renewal Program	+40 mi avg fleet range	18–24 months	MEDIUM	High

Immediate Actions

R1 + R2 + R4 can all be actioned within 12 months using existing WA State legislative frameworks. Combined projected impact: 30,000+ additional clean-vehicle registrations.

Long-Term Resilience

R3 (rural grid MOUs) must begin now. EV density will arrive before infrastructure is ready without proactive inter-agency coordination agreements.

Fleet Renewal Cycle

50,163 Growth-wave vehicles (2016–2020) are entering peak replacement age 2026–2030. Demand surge is structural. Automakers and utilities should prepare fleet renewal programs.

Limitations & Constraints

DATA LIMITATIONS

Missing Range Data (64% of Raw)

179,538 of 280,813 rows had Electric Range = 0 — DOL code for 'not researched'. Analysis uses 101,275 verified records only. PHEVs are disproportionately absent, skewing BEV/PHEV range comparisons.

Registration ≠ Driving Location

A King County-registered EV may commute daily to Snohomish. Charging demand is systematically undercounted outside urban cores in registry-based analysis.

No Demographic / Income Data

Registry contains no income, household, or socioeconomic fields. Equity analysis requires an external ACS Census join at census tract level.

Snapshot Dataset (No Panel)

The dataset captures the current registry state only. Historical deregistrations, retirements, and fleet turnover patterns are not tracked.

METHOD LIMITATIONS

Brand-Level Market Segmentation

Market_Segment assigned at brand level, not by MSRP. Entry-level luxury (BMW 2-Series PHEV) and premium mass-market (Bolt EUV) may be misclassified.

Correlation ≠ Causation

Pearson $r = -0.268$ (Model Year vs Range) confirms association but does not isolate the causal driver — technology improvements, market mix shifts, and DOL reporting changes all contribute.

278 Out-of-WA Coordinates

Records with Latitude < 45.5 or Longitude > -116.9 are likely registry entry errors. Excluded from map visualisations but retained in tabular aggregations.

Next Steps & Closing Summary

FUTURE EXTENSIONS

01 ACS Income Data Join

Join Census ACS income data by census tract → equity-adjusted adoption maps revealing socioeconomic distribution patterns.

02 ARIMA / Prophet Forecasting

Time-series models on monthly registration counts → 2027–2030 adoption projections for infrastructure capacity planning.

03 Charging Gap Analysis

Overlay AFDC public charger locations with registration density → identify under-served corridors, prioritise new deployments.

04 Panel Data Tracking

Request historical DOL snapshots → analyse deregistration rates, fleet turnover, and actual vehicle lifecycle patterns.

05 Grid Capacity Stress-Test

Combine Utility_Complexity_Score with utility published capacity data → infrastructure stress-testing per county.

CLOSING SUMMARY

This capstone delivered a production-grade analytics framework on 280,813 EV registrations — a 5-stage ETL pipeline, 9 engineered features, 3 validated statistical tests (Pearson r, T-Test, ANOVA), and 3 interactive Tableau dashboards. Core finding: Washington State's EV transition has crossed its tipping point. Equitable adoption now requires targeted charging infrastructure in growth counties, mass-market incentive reform, and proactive rural grid coordination before demand arrives.